Name → Varun Chauhan
2401010276

Assignment → 4

SECTION → A

1) Transaction in DBMS:
A transaction is a logical unit of work that consists of multiple database operations (like insert, update, delete) executed as a single unit. It ensure that either all operations are compleat successfully or none are applied, maintaining database consistency.

2) ACID properties:

* Atomicity → All or nothing execution
* Consistency → Database moves from one valid state to another.
* Isolation → Transactions do not interfere with each other.
* Durability → Once committed, changes are permanent.

3) Concurrency control.

It is the mechanism used to to manage multiple transactions executing simultaneously without causing inconsistency. It ensures correctness while improving system performance.

4) Deadlock :-

A deadlock occurs when two or more transactions wait indefinitely for each others resources, and none can proceed.

5] Serializability

It ensures that concurrent execution of transactions gives the same result as if they were executed serially (one after another.)

6] Recovery in DBMS.

Recovery is the process of restoring the database to a consistent state after a failure using logs, undo and redo operations.

7] Checkpoint.

A checkpoint is mechanism that saves the current state of the database, reducing recovery time by avoiding reprocessing the entire log.

SECTION → B

1] Acid with Banking Example

In a money transfer (A → B) :-

* Atomicity → Debit and credit must both happen.
* Consistency → Total balance remains unchanged
* Isolation → No other transaction sees partial update.

* Durability → Once completed, data is permanently stored.

2) Deadlock situation

Eg →
* Transaction T_1 locks Account A, waits for B.
* Transaction T_2 locks Account B, waits for A.

Both wait forever → Deadlock.

3) Serial vs Concurrent Execution.

* Serial Execution → Transactions run one by one
* Concurrent Execution → Multiple transactions run simultaneously.

→ Concurrent is faster but needs control.
→ Serial is safer but slower.

SECTION → C

1) Banking Transfer & Acid.

* Atomicity → Either full transfer or none.
* Consistency → No loss of money.
* Isolation → Other users don't see intermediate state.
* Durability → Final balance persists.

2) Two transactions updating same data.

→ Possible issues :—

* Lost update
* Dirty Read.
* Non - Repeatable Read.

These are concurrency problems solved using locking or isolation levels.

3) System crash During Transaction

i) Check log file.
ii) Undo incomplete transactions.
iii) Redo committed transactions.
iv) Restore consistent database state.